

Module 5: Unit 5.2

Oscillator



Objectives

- ◆ Basic concept of an oscillator
 - ◆ Principles of operation of an oscillator
 - ◆ Condition for Oscillation
 - ◆ Types of Oscillators
 - RC Oscillator
 - LC oscillator
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Introduction

- ◆ Oscillator is an electronic circuit that generates a periodic waveform on its output without an external signal source. It is used to convert dc to ac.
 - ◆ Oscillators are circuits that produce a continuous signal of some type without the need of an input.
 - ◆ These signals serve a variety of purposes.
 - ◆ Communications systems, digital systems (including computers), and test equipment make use of oscillators
 - ◆ An oscillator is a circuit that produces a repetitive signal from a dc voltage.
 - ◆ The feedback oscillator **relies on a positive feedback** of the output to **maintain the oscillations**.
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Types of oscillators

1. RC oscillators

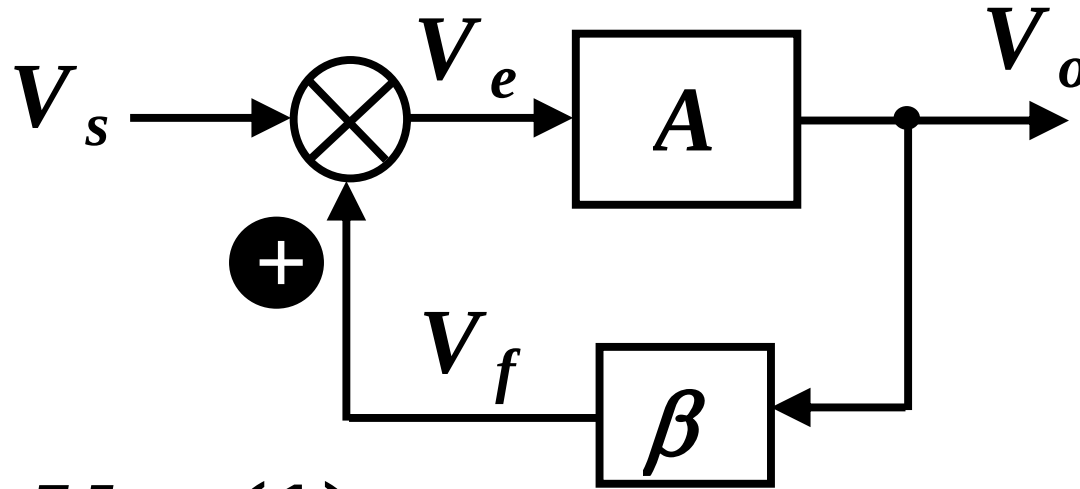
- Wien Bridge
- Phase-Shift

2. LC oscillators

- Hartley
 - Colpitts
 - Crystal
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Basic principles for oscillation

- ◆ An oscillator is an amplifier with positive feedback.



$$V_e = V_s + V_f \quad (1)$$

$$V_f = \beta V_o \quad (2)$$

$$V_o = A V_e = A(V_s + V_f) = A(V_s + \beta V_o) \quad (3)$$

Basic principles for oscillation

$$\begin{aligned}V_o &= AV_e \\&= A(V_s + V_f) = A(V_s + \beta V_o)\end{aligned}$$

$$\begin{aligned}V_o &= AV_s + A\beta V_o \\(1 - A\beta)V_o &= AV_s\end{aligned}$$

◆ The closed loop gain is:

$$A_f \equiv \frac{V_o}{V_s} = \frac{A}{(1 - A\beta)}$$

Basic principles for oscillation

- ◆ In general A and β are functions of frequency and thus may be written as;

$$A_f(s) = \frac{V_o}{V_s}(s) = \frac{A(s)}{1 - A(s)\beta(s)}$$

$A(s)\beta(s)$ is known as **loop gain**

Basic principles for oscillation

- ◆ Writing $T(s) = A(s)\beta(s)$ the loop gain becomes;

$$A_f(s) = \frac{A(s)}{1 - T(s)}$$

- ◆ Replacing s with $j\omega$

$$A_f(j\omega) = \frac{A(j\omega)}{1 - T(j\omega)}$$

- ◆ and $T(j\omega) = A(j\omega)\beta(j\omega)$
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Basic principles for oscillation

- ◆ At a specific frequency f_0

$$T(j\omega_0) = A(j\omega_0)\beta(j\omega_0) = 1$$

- ◆ At this frequency, the closed loop gain;

$$A_f(j\omega_0) = \frac{A(j\omega_0)}{1 - A(j\omega_0)\beta(j\omega_0)}$$

will be infinite, i.e. the circuit will have finite output for zero input signal - oscillation

Basic principles for oscillation

- ◆ Thus, the condition for sinusoidal oscillation of frequency f_0 is;

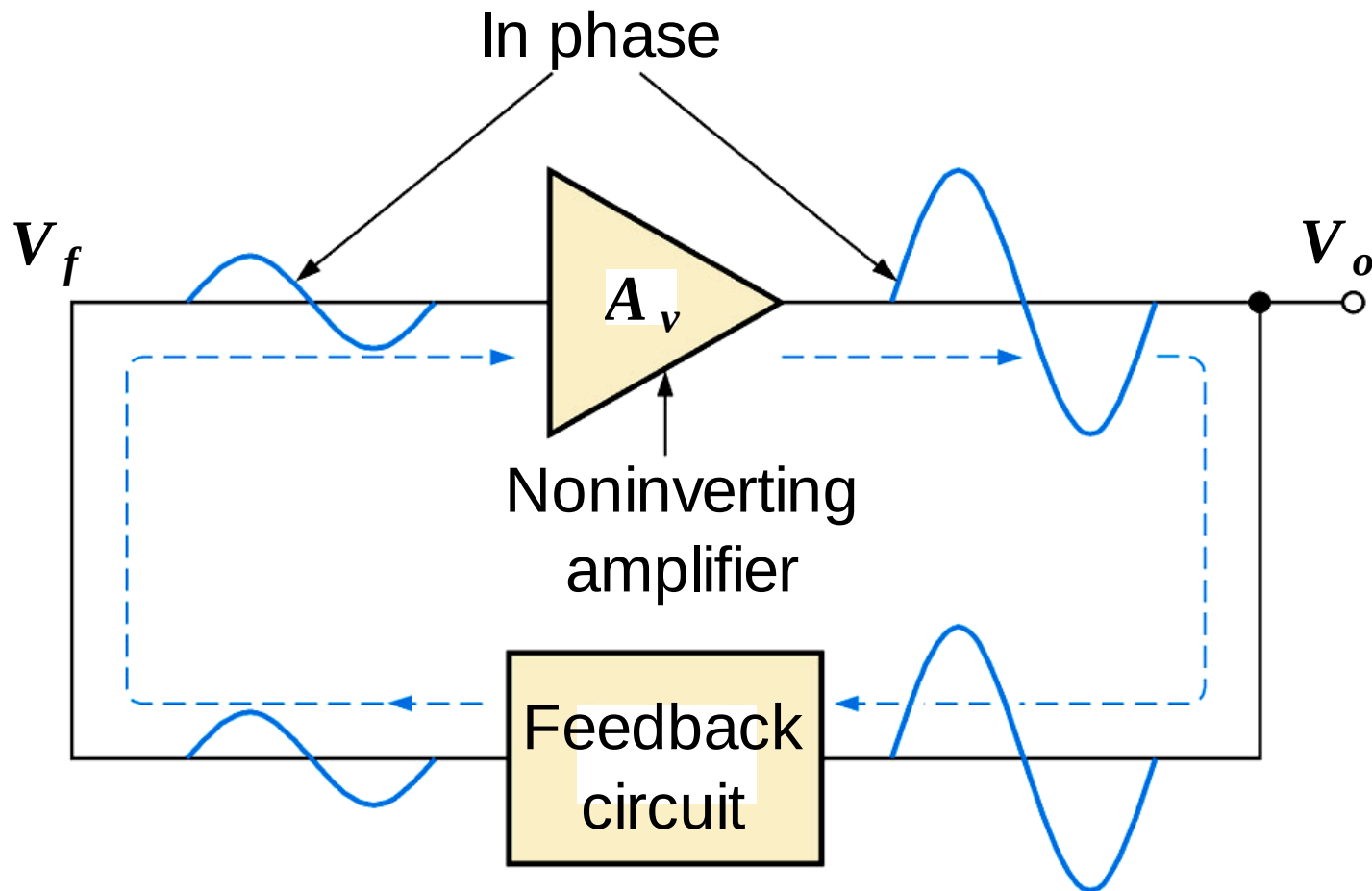
$$A(j\omega_0)\beta(j\omega_0) = 1$$

- ◆ This is known as **Barkhausen criterion**.
 - ◆ The frequency of oscillation is solely determined by the phase characteristic of the feedback loop – the loop oscillates at the frequency for which the phase is zero.
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Basic principles for oscillation

- ◆ The feedback oscillator is widely used for generation of sine wave signals.
 - ◆ The positive (in phase) feedback arrangement maintains the oscillations.
 - ◆ The feedback gain must be kept to unity to keep the output from distorting.
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Basic principles for oscillation



Design Criteria for Oscillators

1. The magnitude of the loop gain must be unity or slightly larger

$$|A\beta| = 1 \quad \text{– Barkhausen criterion}$$

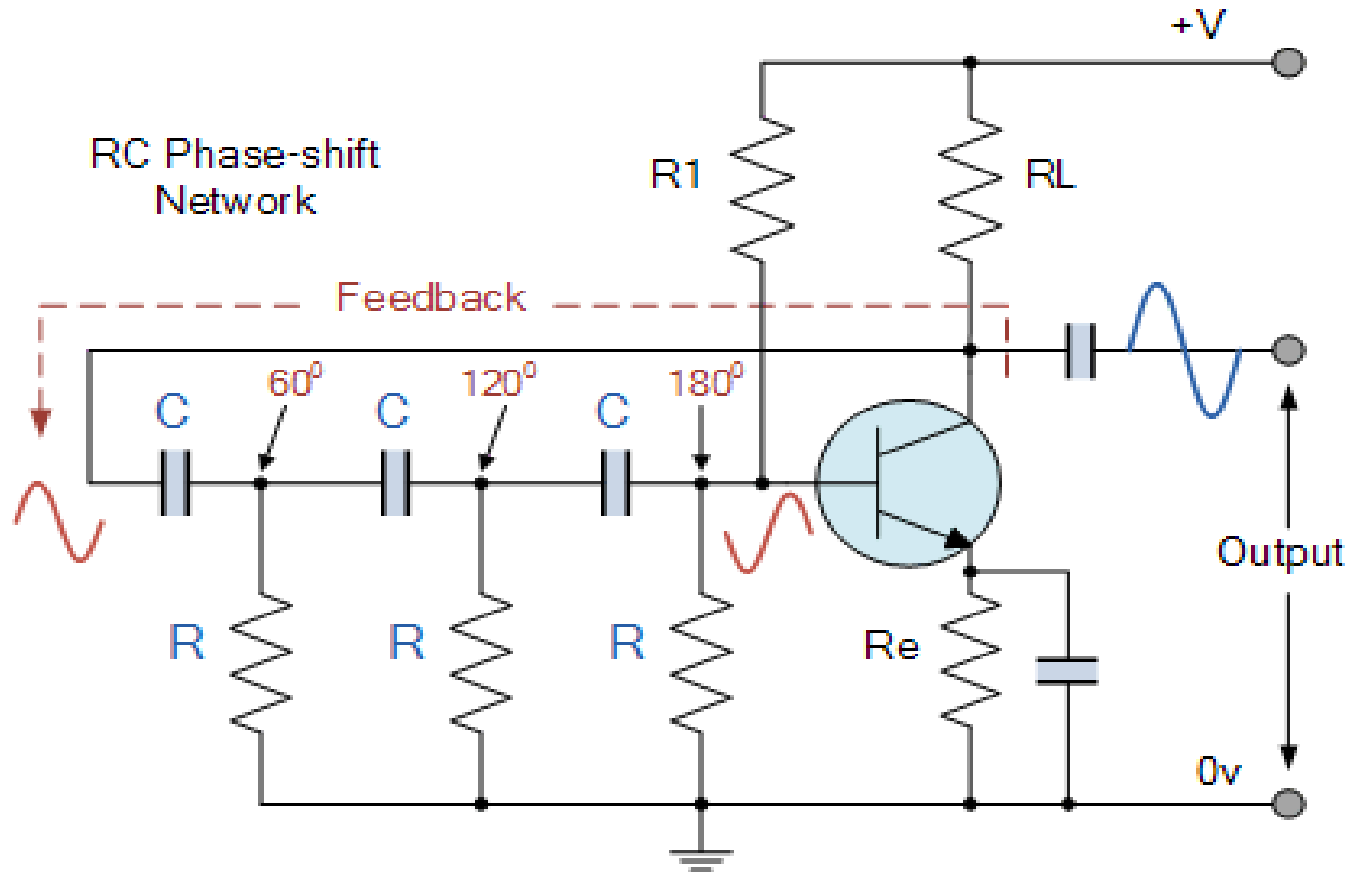
2. Total phase shift, ϕ of the loop gain must be $N \times 360^\circ$ where $N=0, 1, 2, \dots$

RC Phase-Shift Oscillator

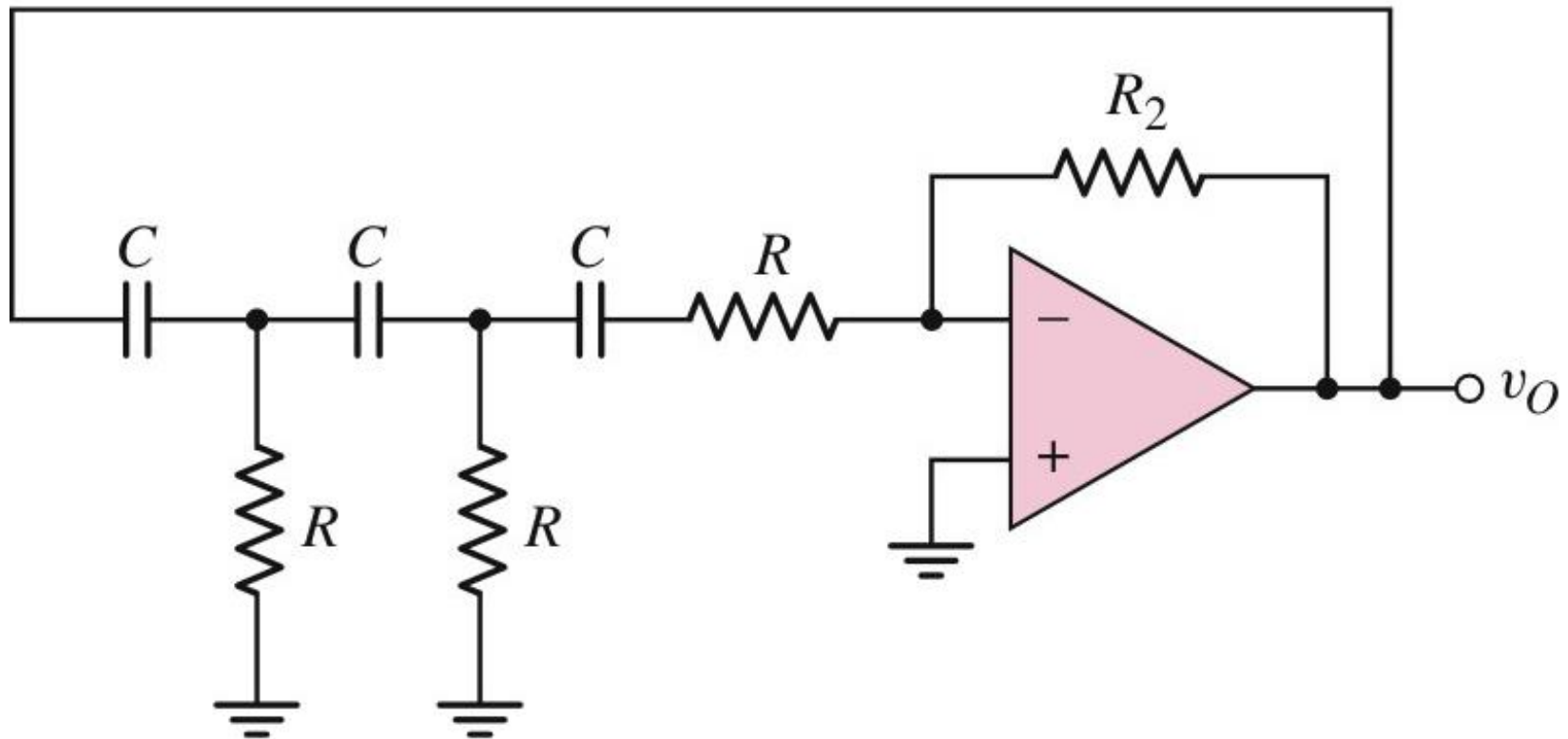
- ◆ The phase shift oscillator utilizes **three RC circuits to provide 180° phase shift** that when coupled with the 180° of the op-amp itself provides the necessary feedback to sustain oscillations.
- ◆ The **gain must be at least 29 / 42.5** to maintain the oscillations.
- ◆ The frequency of resonance for the this type is similar to RC circuit oscillator using BJT

$$f_r = \frac{1}{2\pi RC(\sqrt{6 + 4R_c / R})}$$

Phase-Shift Oscillator



Phase-Shift Oscillator



$$f_o = \frac{1}{2\pi\sqrt{6}RC} \quad \frac{R_2}{R} = 29$$

The gain must be at least 29 to maintain the oscillations

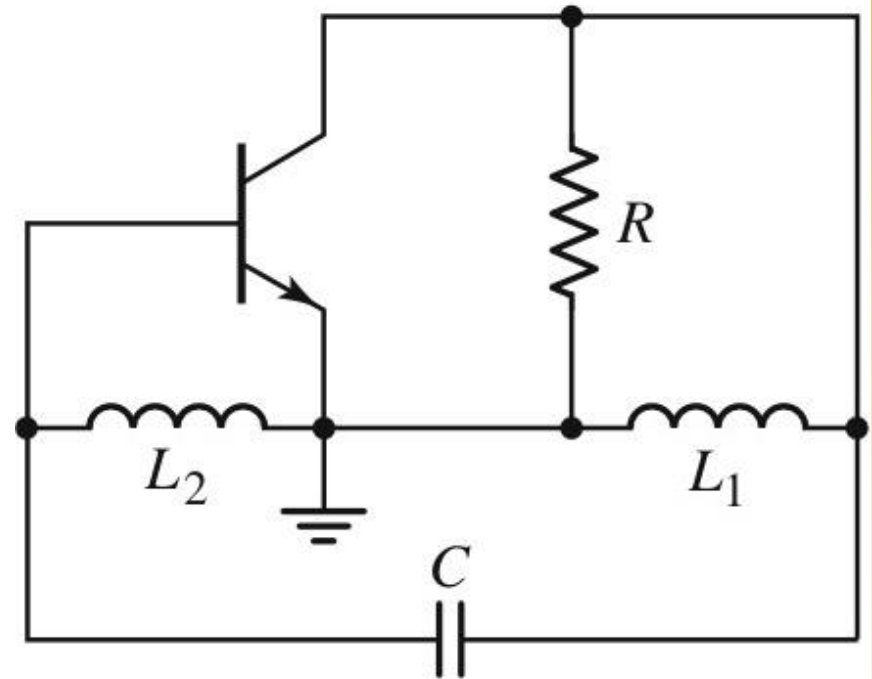
LC Oscillators

- ◆ Use transistors and LC tuned circuits or crystals in their feedback network.
 - ◆ For hundreds of kHz to hundreds of MHz frequency range.
 - ◆ Colpitts, Hartley and crystal oscillator.
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Hartley Oscillator

- ◆ The feedback network of the Hartley oscillator uses *tapped inductors* (L_1 and L_2) and *a single capacitor* C .
- ◆ The frequency of oscillation:

$$\omega_o = \frac{1}{\sqrt{(L_1 + L_2)C}}$$

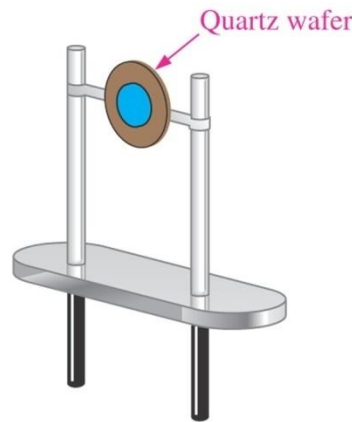


Crystal Oscillator

- ◆ Most communications and digital applications require the use of oscillators with **extremely stable output**. Crystal oscillators are invented to overcome the **output fluctuation** experienced by conventional oscillators.
- ◆ Crystals used in electronic applications consist of a quartz wafer held between two metal plates and housed in a package as shown in Fig. 9 (a) and (b).



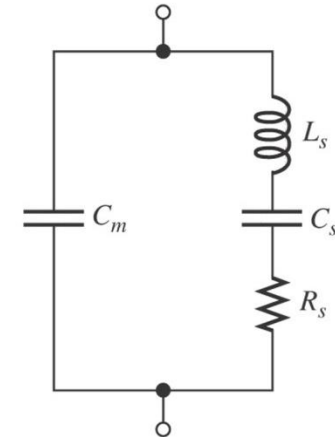
(a) Typical packaged crystal



(b) Basic construction (without case)



(c) Symbol



(d) Electrical equivalent

Crystal Oscillator

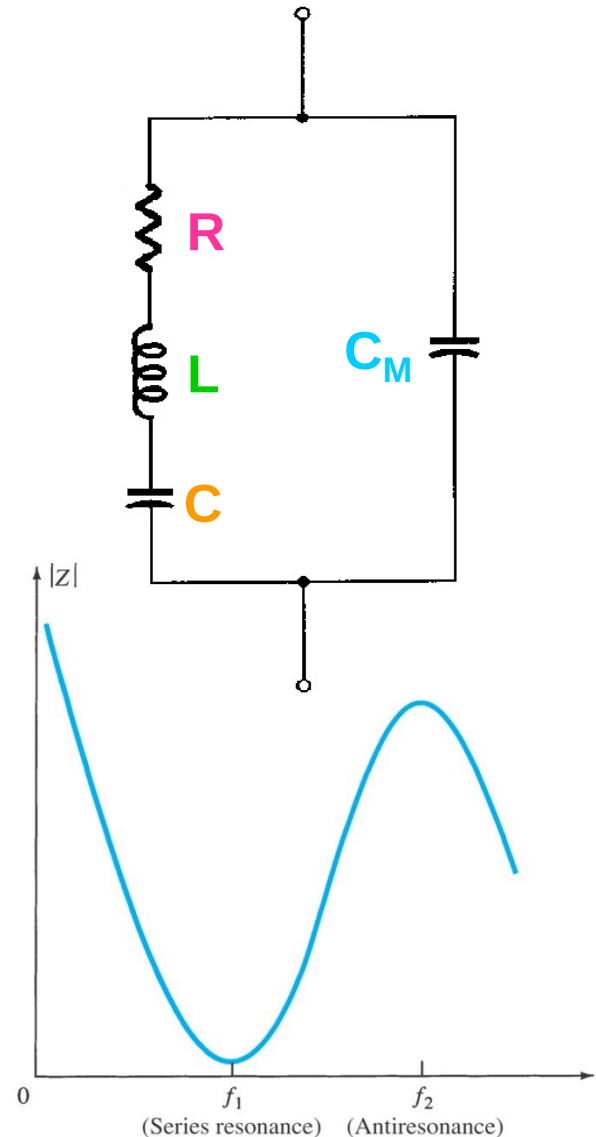
◆ Piezoelectric Effect

- The quartz crystal is made of silicon oxide (SiO_2) and exhibits a property called the *piezoelectric*
- When a changing an alternating voltage is applied across the crystal, it vibrates at the frequency of the applied voltage. In the other word, the frequency of the applied ac voltage is equal to the natural resonant frequency of the crystal.
- The thinner the crystal, higher its frequency of vibration. This phenomenon is called piezoelectric effect.

Crystal Oscillator

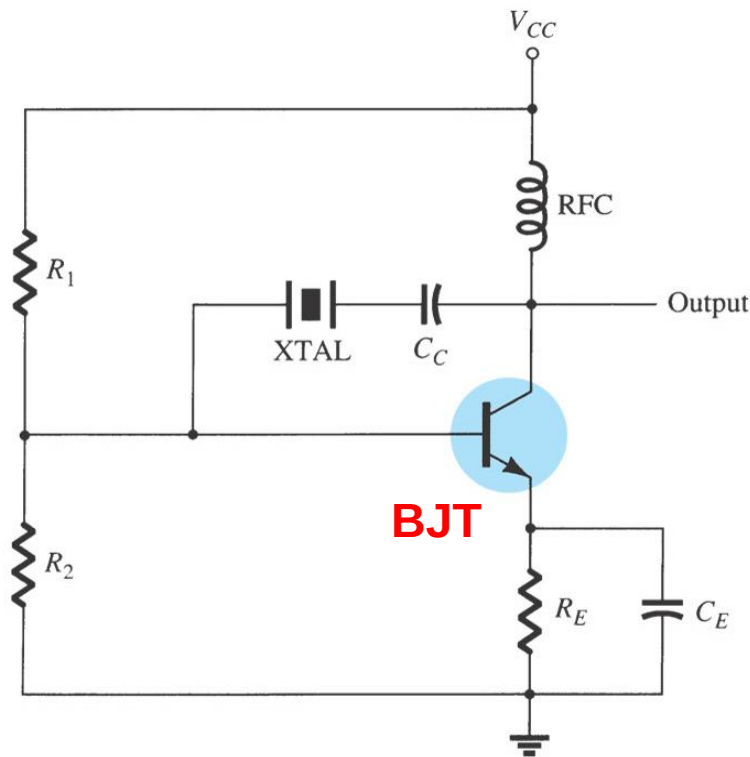
◆ Characteristic of Quartz Crystal

- The crystal can have two resonant frequencies;
- One is the series resonance frequency f_1 which occurs when $X_L = X_C$. At this frequency, crystal offers a very low impedance to the external circuit where $Z = R$.
- The other is the parallel resonance (or antiresonance) frequency f_2 which occurs when reactance of the series leg equals the reactance of C_M . At this frequency, crystal offers a very high impedance to the external circuit

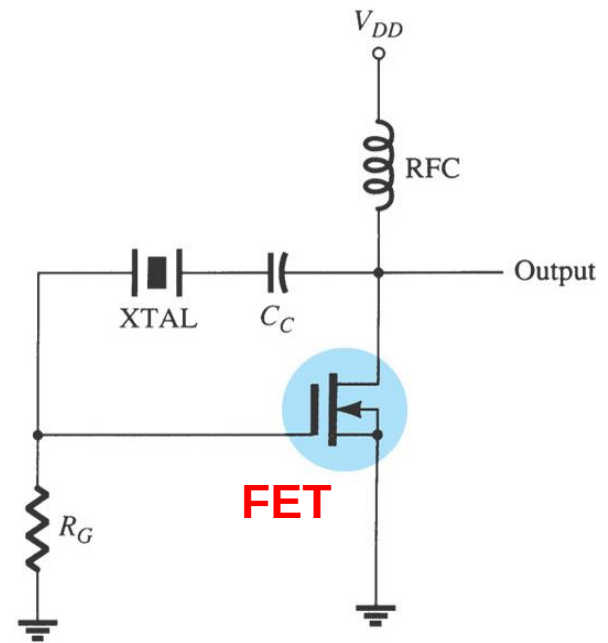


Crystal Oscillator

- ◆ The crystal is connected as a series element in the feedback path from collector to the base so that it is excited in the series-resonance mode



(a)



(b)

Crystal Oscillator

- ◆ Since, in series resonance, crystal impedance is the smallest that causes the crystal provides the largest positive feedback.
- ◆ Resistors R_1 , R_2 , and R_E provide a voltage-divider stabilized dc bias circuit. Capacitor C_E provides ac bypass of the emitter resistor, R_E to avoid degeneration.
- ◆ The RFC coil provides dc collector load and also prevents any ac signal from entering the dc supply.
- ◆ The coupling capacitor C_C has negligible reactance at circuit operating frequency but blocks any dc flow between collector and base.
- ◆ The oscillation frequency equals the series-resonance frequency of the crystal and is given by:

$$f_o = \frac{1}{2\pi\sqrt{LC_C}}$$